Dynamic Adaptive Region Transformer for Tiny-Object Detection in Remote Sensing

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***Abstract*— Despite detection Transformer (DETR)-like methods having improved end-to-end detection capabilities, they fundamentally struggle with the uniform processing of entire flattened feature maps, causing queries to attend to irrelevant regions. This results in redundant attention patterns, leading to computational burdens and inefficiencies in the detection of tiny objects in remote sensing imagery across dense and sparse settings. To address these limitations, we present a lightweight Transformer-based encoder–decoder architecture called dynamic adaptive region Transformer (DART). Specifically, density adaptive Transformer (DAT) employs an adaptive-region attention (ARA) mechanism that dynamically generates content-aware spatial regions rooted in feature density and semantic relevance. This strategy prioritizes computational resources on semantically rich areas while minimizing focus on irrelevant background regions. Region-aware decoder (RAD) incorporates a masked region-aware cross-attention (MRA) mechanism, where queries interact exclusively with the adaptive masked regions generated by the encoder, thereby reducing redundant focus on overlapping or irrelevant areas. Meanwhile, a query diversity loss is introduced to penalize overlapping attention patterns among queries, encouraging each query to focus on distinct and complementary regions. By adapting to data density and directing queries to essential areas within the image, DART enhances feature extraction and object localization for various-sized objects in dense and sparse settings. Experimental results demonstrate that DART achieves state-of-the-art performance on AI-TOD, DOTA-v2.0, and LEVIR-Ship benchmarks and exhibits strong generalization capabilities on DIOR while using only 13 million parameters and a computational cost of 68 GFLOPs.**

***Index Terms*— Content-aware spatial regions, remote sensing images, tiny-object detection, Transformer.**

I. Introduction

TINY objects are ubiquitous in remote sensing imagery, enabling numerous applications such as vehicle detection [1] and marine surveillance [2]. While object detection has seen significant advancements, including methods like faster R-CNN [3] and FCOS [4], the focus has largely been on larger objects that benefit from distinct features. Detecting tiny objects, often smaller than 16 pixels, presents unique challenges, particularly when they appear across varying densities due to their minimal visual data. In densely populated areas, such as parking lots where cars may occupy over 50% of the image, object overlap complicates localization, leading to missed detections. Conversely, sparsely distributed objects, like very tiny ships on the open sea occupying less than 1% of the image, lack contextual cues, resulting in increased false positives (FPs). Moreover, scale disparity obscures tiny objects, complicating their detection by larger ones, as illustrated in Fig. 1.



Fig. 1. Performance of DART on irregular object distribution (i.e., dense and sparse) and scale variation. The image, taken from the DOTA-v2.0 validation set, illustrates how DART effectively manages varying object scales and uneven distributions in remote sensing while making efficient use of computational resources. The green, blue, and red boxes represent TPs, FPs, and FNs, respectively.

Recently, many efforts have been devoted to tiny-object detection in remote sensing. A primary strategy involves enhancing context modeling within the backbone to distinguish inconspicuous tiny objects from their backgrounds [5], [6]. Complementing this, multiscale feature fusion strategies further amplify foreground features while minimizing

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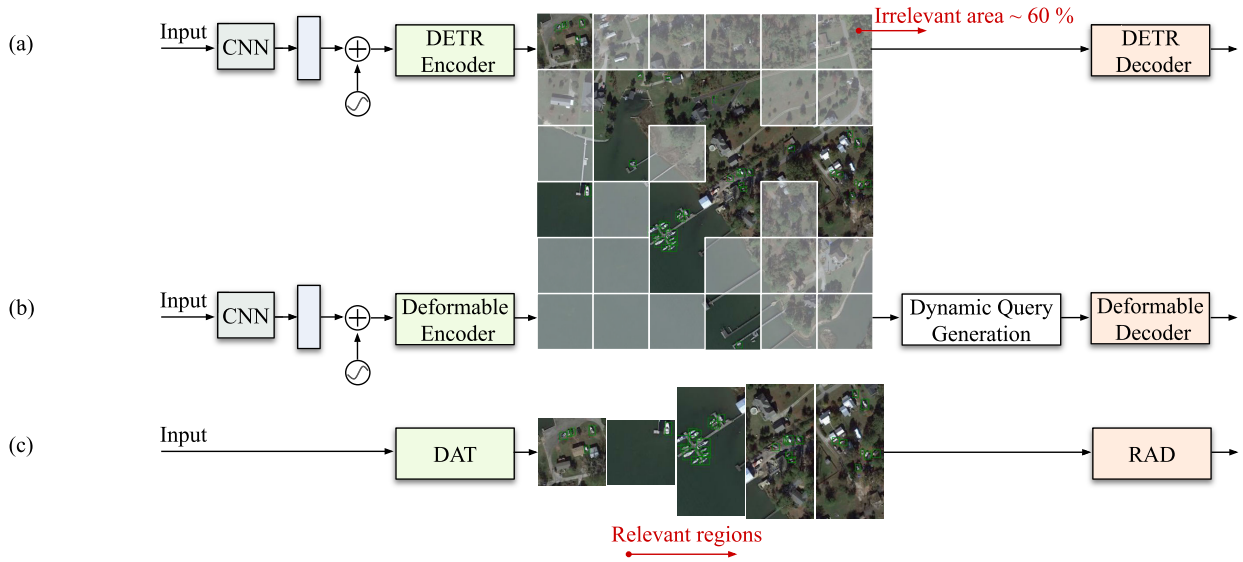


Fig. 2. Schematic of DETR-based approaches. (a) DETR processes the entire feature map uniformly, causing attention redundancy and inefficient focus on irrelevant areas. (b) DQ-DETR aims to mitigate this by using a deformable encoder, but it still struggles with inefficient attention allocation to irrelevant regions. (c) DART introduces the DAT to replace the CNN-based backbone and Transformer encoder in DETR. By focusing attention on the most relevant areas, DAT adapts to spatial data density more effectively, reducing redundancy and enhancing object localization, particularly in contexts with varying densities, such as dense vehicle and maritime environments.

background noise by capturing information across various scales [7], [8]. In contrast, some efforts focus on designing better training strategies to selectively train objects based on their scale and background complexity [9], [10], while others focus on designing label assignment strategies to assign high-quality anchors to the ground truth boxes of various-sized objects [11], [12]. While convolutional neural network (CNN)-based approaches have improved tiny-object detection performance to some extent, their reliance on manually designed components, such as nonmaximum suppression (NMS) and anchor generation, limits their potential for end-to-end optimization.

Detection Transformer (DETR) [13] streamlines object detection by eliminating hand-crafted components, and instead leveraging a Transformer architecture to reason about global image relationships and directly output predictions. However, its slow convergence attributed to cross-attention, Hungarian matching instability, and object queries, combined with its struggle with varying object densities, has spurred various modifications. Encoder-only DETR [14] eliminates the decoder to accelerate training, while DN-DETR [15] employs denoising training to simplify bipartite graph matching. Others enhance decoder queries by directly manipulating their representation, with deformable DETR [16] using 2-D reference points, and efficient DETR [17] selecting Top-K positions to improve convergence and small-object detection. DQ-DETR [18] enhances tiny-object detection in remote sensing by dynamically varying query counts from 300 to 1500, adapting to different densities.

This work argues that current DETR-like methods struggle with uniform processing of the entire feature map, causing queries to attend to irrelevant areas, resulting in attention redundancy, as demonstrated in Fig. 2. For example, in dense vehicle scenarios like parking lots, cross-attention mechanisms force queries to evaluate all spatial positions uniformly, ignoring the underlying density of the feature map, and causing feature confusion when queries redundantly focus on similar or irrelevant regions. Similarly, in maritime applications, the global attention approach wastes significant computational resources by forcing queries to process vast empty oceanic regions to detect sparse ship distributions. While some approaches aim to mitigate attention redundancy by enhancing query optimization, such as pruning redundant queries [19] employing query routing [20], or diversifying query content [21], these methods often fail to address the core issue of indiscriminate attention applied to irrelevant spatial regions. This emphasizes the necessity for a more focused strategy that adapts to data density and directs queries to essential areas within the image to facilitate the extraction of distinguishing features and improve performance in context-sensitive analysis and accurate object localization.

Based on these findings, we present a lightweight Transformer-based encoder–decoder architecture called dynamic adaptive region Transformer (DART) for tiny-object detection in remote sensing. DART features a density adaptive Transformer (DAT) and region-aware decoder (RAD). DAT employs an adaptive-region attention (ARA) mechanism that dynamically adjusts feature processing to optimize attention allocation and enhance detection performance. Specifically, it first generates adaptive sampling points that evolve based on the image's inherent feature density, concentrating on regions of higher richness to focus on semantically relevant areas. Next, a relational framework is established among the points based on similarity and proximity, grouping spatially close points or those with shared features to facilitate coherent and contextually aligned attention patterns. Finally, these relational structures are transformed into continuous dynamic, content-aware spatial regions to compute attention, ensuring that computational resources are focused on semantically rich areas while ignoring irrelevant background regions. Once the regions are formed, RAD isolates masked regions to

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preserve content while suppressing other locations, generating regions with the highest object concentration. RAD then employs a masked region-aware cross-attention (MRA) mechanism, where queries interact only with the masked adaptive regions generated by the encoder. By computing queries over distinct regions, it restricts queries to relevant spatial areas, minimizing redundant focus on overlapping or irrelevant regions while maximizing computational efficiency. Additionally, query diversity loss is introduced to penalize overlapping attention patterns among queries, encouraging each query to focus on distinct and complementary regions. This ensures better feature discrimination and improves the model's ability to detect objects of various sizes across varying densities in intricate scenes.

The contributions of this article are summarized as follows.

1) We introduce DART, a lightweight Transformer-based encoder–decoder architecture that achieves state-of-the-art performance on tiny-object detection benchmarks while using only 13 million parameters and a computational cost of 68 GFLOPs.
2) DAT employs an ARA mechanism that varies from fixed grid-based attention by forming spatial regions based on feature density and semantic relevance, enabling targeted computation for efficient feature extraction of tiny objects in complex remote sensing scenes.
3) RAD replaces cross-attention in DETR with an MRA mechanism that restricts queries to interact only with encoder-derived adaptive masked regions, ensuring that each query focuses on distinct and meaningful spatial areas and minimizing attention redundancy in both dense and sparse scenarios.
4) A novel query diversity loss is introduced to enhance the one-to-one matching strategy by penalizing overlapping attention patterns, encouraging exploration of distinct regions, and improving the discrimination of visually similar or closely packed objects.

II. Related Work

A. Tiny-Object Detection

CNN-based tiny-object detection strategies in remote sensing can be broadly categorized into context modeling within the backbone, multiscale feature fusion, improved training strategies, and label assignment methods.

To tackle the challenges of tiny-object detection, some methods enhance context within the backbone network. Collaborative backbone [22] utilizes spatial–channel reconstruction convolution for shallow feature extraction, whereas global-context backbone [23] employs global attention pooling to efficiently capture contextual relationships with reduced complexity. Additionally, Huang et al. [6] developed a lightweight backbone using XYW-conv and XYW-attention to enhance contrast for tiny objects, but requires 220.2 GFLOPs of computation. Multiscale feature fusion methods further improve detection networks by capturing information at various scales. Liu et al. [24] enhanced feature pyramid networks (FPNs) by increasing the receptive field of convolutional kernels to enhance contextual awareness, while Xiao et al. [25] improved feature correlation in FPNs to preserve small-target details. DNTR with denoising FPNs [8] further improves feature extraction through specialized encoders, but it only reaches 11.9 FPS and demands a significant 167.5 GFLOPs. However, these convolutional layer methods often struggle with long-range dependencies. Additionally, inadequate feature weighting can impair performance, while processing multiple feature maps raises computational demands.

In contrast, some methods emphasize detecting both tiny and large objects by considering scale and background. SNIP [9] optimizes detection performance by selectively training on specific scale ranges, while Sig-NMS [26] reduces the likelihood of missing small targets by adjusting their scores based on overlap with larger objects. Wang et al. [10] classified images by background complexity and trained separate networks for each category to further improve accuracy for tiny objects. While others focus on developing label assignment strategies to ensure accurate anchor assignment for tiny objects. Fu et al. [27] introduced adaptive dynamic anchor mining to mitigate distribution bias between positive and negative samples. Zhang et al. [28] proposed a Gaussian assignment strategy that models bounding boxes as 2-D Gaussian distributions to enhance overall accuracy. Additionally, Shi et al. [29] developed a similarity distance metric to refine anchor assignment for tiny objects by considering bounding-box location and shape. However, concentrating on a single scale or object placement can hinder generalization, resulting in degraded performance with sparse objects after training on dense regions. Conversely, even with complex assignments, objects with dense clustering may bias anchors toward larger or overlapping objects, complicating the identification of smaller, scattered ones.

B. Transformer in Tiny-Object Detection

Recent advances in tiny-object detection for remote sensing imagery leverage self-attention in CNN-based detection framework (i.e., backbone, neck, and detection head) to capture better local–global contexts and long-range dependencies.

Zhang et al. [30] enhanced CSPDarknet53 backbone by using a self-attention mechanism and convolutional layers, whereas Li et al. [31] further improved it by using global and local attention to optimize feature extraction. Similarly, Huo et al. [32] enhanced EfficientNetv2-s with a 2L-Transformer block that combines local and global self-attention for improved contextual information. LGFormer [5] utilizes a local and global Transformer block to capture local token correlations and long-range dependencies, further improving tiny-object detection. However, this comes at the cost of 38.71 million parameters and a processing speed of only 15.5 FPS. To enhance local–global context across multiple layers, some approaches incorporate self-attention within FPN. For example, Zhang et al. [7] developed an efficient attention mechanism that mimics self-attention while reducing computational complexity. Similarly, Zhao et al. [33] introduced an attention-based foreground enhancement module that amplifies features and reduces background noise to further enhance the detection of dense tiny objects. In contrast,

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Huo et al. [34] replaced FPNs with a Swin Transformer-based neck to better integrate multiscale features for very small objects. Conversely, other methods focus on enhancing the detection head. For example, Liu et al. [8] introduced the Trans R-CNN detector, which applies self-attention in the detection head to capture global context within a region of interest (RoI). Similarly, Chen et al. [2] designed a cross-stage multihead attention module that expands the receptive field to enhance the feature discrimination for detecting tiny ships. While self-attention-based detectors enhance long-range dependencies, their reliance on predefined grid-based uniform attention can lead to inefficiencies in feature aggregation due to misalignment with image density. In contrast, our ARA mechanism improves detection by dynamically adjusting spatial attention to the scene's characteristics, achieving state-of-the-art performance at 45 FPS with 13 million parameters and 68 GFLOPs.

C. DETR and Its Variants in Tiny-Object Detection

DETR [13] is an end-to-end object detector that utilizes Transformers without relying on manually designed components. The Transformer encoder extracts instance-level features from an image, while the Transformer decoder employs a set of learnable queries to probe and pool these features. Although it achieves performance comparable to Faster R-CNN [3], it faces challenges with slow convergence and difficulties in handling varying object densities. Numerous follow-up studies have explored various strategies to address this issue, including the design of cross-self attention, optimizations of the loss function, enhancements to learnable queries, and improved detection across varying densities.

To tackle the inefficiency of DETR's decoder cross-attention, dynamic DETR [35] refine it by focusing on ROI, while other approaches, like encoder-only DETR [14] discard it entirely, opting for direct prediction from encoder features to improve training speed. While some attribute DETR's slow convergence to the Hungarian loss. To analyze this, Sun et al. [14] employed a teacher model for predefined assignments. They found that although this approach facilitated early convergence, it was not the primary factor contributing to the observed results. Building on this, DN-DETR [15] adds a denoising loss alongside the Hungarian loss to further accelerate training. DINO [36] further enhances one-to-one matching with contrastive denoising training using positive and negative samples to prevent duplicates. Another series of works enhances DETR's decoder queries by directly manipulating their representation to accelerate convergence. Deformable DETR [16] uses 2-D reference points as queries and performs deformable cross-attention. Conditional DETR [37] decouples attention and generates positional queries from reference coordinates. Efficient DETR [17] uses a dense prediction module to select top-K positions as object queries. DAB-DETR [38] uses 4-D box coordinates as queries and updates this layer by layer. Building on this foundation, Zhao et al. introduced RT-DETR [39], the first real-time, end-to-end object detector to achieve state-of-the-art performance. It utilizes attention-based intrascale feature interaction and CNN-based cross-scale feature fusion to enhance inference speed and accuracy.

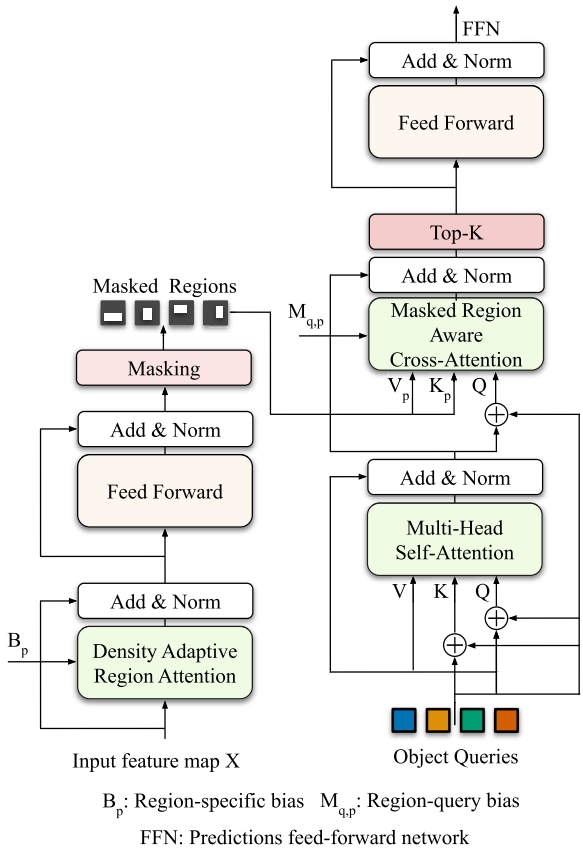


Fig. 3. Architecture of DART, featuring (left) DAT, which optimizes attention allocation for varying object densities, and (right) RAD, which employs masked cross-attention to focus on relevant regions.

In remote sensing, Han et al. [40] introduced CI_DETR, a novel Transformer-based object detection method that enhances the original DETR architecture by incorporating a multilevel feature fusion module, which enables the network to learn spatially adaptive features at different levels while preserving relevant information. Ma et al. [41] introduced QETR, which innovatively employs a query alignment module to refine local object information within a limited attention range, along with a scale controller to preserve object shape across decoder blocks. Mo and Zhu [42] designed dynamic anchor object queries to enable the network to detect a wider range of multiscale and densely distributed objects more accurately. Zhong et al. [43] introduced PSWP-DETR, which leverages PredatorConv to enhance the backbone's ability to adapt to and capture complex geometric features of tiny objects in remote sensing images by learning horizontal and vertical deformation offsets. To further improve tiny-object detection across varying densities, Guo et al. [44] proposed the corresponding regional attention upsample and pooling operators, which use local cross-attention between scales to address foreground-background confusion and enhance feature focus for dense tiny objects. Hu et al. [45] proposed EMO2-DETR, which enhances object detection by employing a reassigned bipartite graph matching strategy to categorize samples, an ignored sample predicted head to identify high-quality negatives, and a reassigned Hungarian loss to focus the model on accurately detecting small objects in dense settings.

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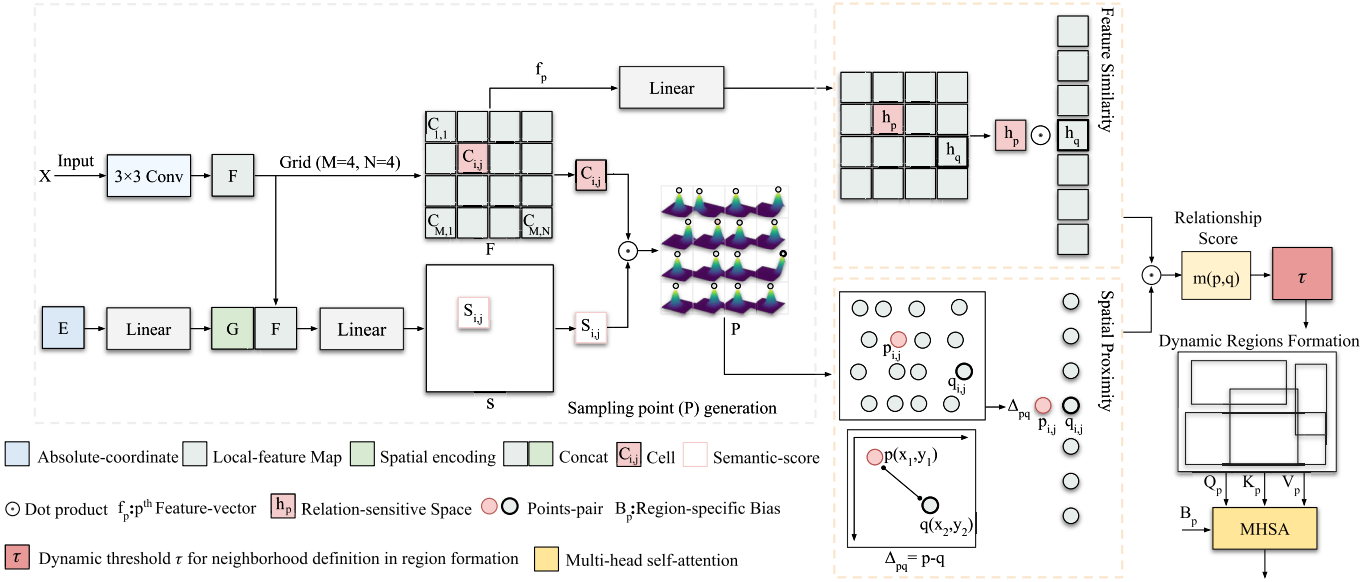


Fig. 4. Architecture of ARA in the DAT. It begins with feature extraction through convolution and linear layers to generate local features and spatial encoding. Adaptive sampling points are then dynamically generated based on the feature density of the image, focusing attention on semantically rich regions. A relational framework is established between the sampling points based on spatial proximity and feature similarity. These relationships are used to form dynamic regions of interest, which are refined by applying a threshold to concentrate on the most relevant areas.

In contrast to these methods that rely on a fixed number of queries, Huang et al. [18] introduced DQ-DETR, which uses a separately trained categorical counting module (CCM) to classify object counts into four levels: fewer than 10, 10–100, 100–500, and over 500, allowing for adaptive query adjustment to different densities at the cost of 58.7 million parameters. However, DETR-like methods face attention redundancy, causing inefficiencies in remote sensing imagery. To address this, we introduce RAD, which uses masked regions for attention computation, directing queries to relevant areas and reducing computational burdens.

III. Methodology

Fig. 3 illustrates DART, a lightweight Transformer-based encoder–decoder framework designed for tiny-object detection. In this framework, DAT uses the ARA mechanism to dynamically generate spatial regions that prioritize computational resources for semantically rich areas, while RAD employs the MRA mechanism to confine queries to relevant regions for attention computation. Furthermore, a query diversity loss promotes distinct focus among queries, enabling them to attend only to complementary regions.

A. Density Adaptive Transformer

1) Architecture of DAT: This article presents a new Vision Transformer named DAT that enhances efficiency by substituting the standard multihead self-attention (MHSA) module in a Transformer block with ARA, which confines self-attention computations to dynamic, content-aware spatial regions, as illustrated in Fig. 4. DAT is specifically designed to replace the traditional CNN backbone and Transformer encoder structure in DETR-based models, thereby improving performance and reducing computational overhead. Given an input image $I \in \mathbb{R}^{H_0 \times W_0 \times C_0}$, DAT processes it through the nth Transformer layers, where $n = 4$. The hyperparameters are $C = 32$, a layer configuration of $[2, 2, 4, 2]$, and attention heads of $[3, 6, 12, 24]$. Fig. 3 (left) illustrates DAT consists of an ARA module followed by a two-layer MLP with GELU activation [see (1) and (2)]. Each ARA module and MLP is preceded by a LayerNorm (LN) layer, and a residual connection is applied after each module. For hierarchical feature learning, the DAT progressively reduces spatial resolution while enhancing channel dimensions at each stage, utilizing a downsampling (DS) block that includes a 3×3 convolution followed by 1×1 convolution. The hierarchical design allows for modeling at diverse scales and maintains linear computational complexity relative to image size. The relationships are given by $H_i, W_i = (H_0/2^i), (W_0/2^i)$, and $C_i = C_0 \cdot 2^i$, where i indicates the stage number, and H_0, W_0, C_0 are the initial dimensions

$$\hat{z}^l = \mathrm{ARA}\left(\mathrm{LN}\left(z^{l-1}\right)\right) + z^{l-1} \tag{1}$$

$$z^l = \mathrm{MLP}\left(\mathrm{LN}\left(\hat{z}^l\right)\right) + \hat{z}^l \tag{2}$$

where $\hat{z}^l$ and z^l denote the output features of the ARA module and the MLP module for lth encoder block, respectively.

2) ARA Mechanism: Since the introduction of the ViT [46], subsequent improvements have focused on enhancing attention mechanisms for greater efficiency. The ViT uses self-attention mechanisms to capture global contextual relationships among image patches. However, their reliance on a fixed patch size throughout the network limits computational efficiency. Building on this foundation, the Swin Transformer [47] adopts a nonoverlapping window approach, shifting the position of windows alternately in consecutive Transformer blocks to facilitate interaction between different regions of the input image. This mechanism allows the model to capture both local and global contexts effectively, but it remains constrained by fixed window sizes. Regional-to-local attention [48] enhances

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this by employing a hierarchical tokenization method that extracts regional tokens from large patches (e.g., 28×28) and local tokens from smaller patches (e.g., 4×4). It utilizes an additional set of regional tokens that perform self-attention specifically on these regional tokens to gather global information, yet it is still limited by predefined patch sizes. These rigid structures struggle with inefficiencies in feature aggregation due to the arbitrary alignment of grid cells with irregular object distributions. This misalignment often leads to oversampling regions of low semantic importance while undersampling areas of higher relevance. Consequently, both attention mechanisms fail to effectively capture the fine-grained and variable spatial relationships within the image. On the other hand, deformable attention [16] utilizes a grid of reference points on the feature map, with an offset network computing spatial offsets based on query features. This allows attention to focus on important regions, while bilinear interpolation samples feature at these deformed positions as keys and values. However, it is constrained by a limited number of sampling keys and the lack of key sharing among query tokens. Additionally, its attention mechanism relies on individually learned linear projections, which struggle to capture comprehensive contextual information across varying spatial distributions.

In contrast, ARA adopts a more adaptive approach by dynamically generating sampling points based on the feature density of the image. This flexibility allows ARA to focus on the most relevant areas without relying on fixed window sizes or predetermined token structures. While window attention and regional-to-local attention have their mechanisms for inter-region interaction, ARA establishes direct relationships among attention points based on spatial proximity and feature similarity, resulting in truly adaptable windows of varying sizes. Similarly, unlike deformable attention, which relies on a fixed grid of reference points and computes spatial offsets, ARA adapts its sampling points in real time. This dynamic process effectively captures fine-grained spatial relationships and optimizes attention distribution, resulting in a coherent representation of the visual data. Specifically, ARA operates in three sequential phases. First, it generates an adaptive set of sampling points, where the spatial layout is not predetermined but instead evolves according to the inherent feature density of the image. By employing a content-dependent approach, the sampling points are concentrated in regions exhibiting higher feature richness, ensuring that the model focuses on semantically relevant areas and avoids the inefficiencies of a fixed grid. Second, a relational framework is established among these attention points, which models their interdependencies based on both feature similarity and spatial proximity. This step ensures that points that are spatially close or share similar features are grouped, enabling coherent and contextually aligned attention patterns. Finally, these relational structures are transformed into continuous spatial regions, which are subsequently used in downstream attention mechanisms. By using these adaptive regions, the model can more effectively aggregate features that reflect the nonuniform spatial nature of remote sensing data.

3) Adaptive Sampling Point Generation: Given an input feature map $X \in \mathbb{R}^{H\times W\times C}$, we first extract local features to capture spatially localized patterns and structural information. This allows the model to focus on fine-grained details in the feature map, defined as F

$$F = \sigma(W_k(X)), \quad F \in \mathbb{R}^{H\times W\times C'} \tag{3}$$

where W_k denotes the convolution of size 3×3, $C' = C/4$, and σ denotes the ReLU.

Concurrently, we encode the absolute spatial location of each pixel (i, j) using a learnable normalized coordinate representation [see (4)]. The resulting spatial encoding G is then fused with local features F to create the semantic scores S [see (5)]. These scores capture fine-grained local visual features and broader spatial context, quantifying the semantic relevance of each location and guiding the adaptive point sampling process to focus on the most informative regions

$$G = \text{linear}(E), \quad G \in \mathbb{R}^{H\times W\times D} \tag{4}$$

$$S = \sigma(\text{linear}[F\|G]), \quad S \in \mathbb{R}^{H\times W\times 1} \tag{5}$$

where $E = [(i/H), (j/W)] \in \mathbb{R}^{H\times W\times 2}$ is a fixed coordinate grid containing normalized positions for each spatial location (i, j). (H, W) denotes the height and width of the feature map X and $\|$ indicates concatenation.

To generate the adaptive sampling points, local feature map F is then partitioned into a structured set of $(M \times N)$ cells $C_{i,j}$, imposing a grid-like framework. The semantic scores S are subsequently used to determine the location of the sampling point by calculating a weighted average of the spatial coordinates of all locations within the cell using (6). This allows sampling points to be drawn toward the most semantically relevant areas within each cell, resulting in a set of adaptive sampling points, denoted as $P = \{p_{i,j}\}_{i=1,\,j=1}^{M,\;N}$

$$p_{i,j} = \sum_{h,w\in C_{i,j}} h, w \cdot \text{softmax}\left(S_{h,w}\right), \quad p_{i,j} \in \mathbb{R}^2 \tag{6}$$

where $M \times N = (H/16) \times (W/16)$ and h, w indicates the pixel values of cell $C_{i,j}$.

4) Region Formation: To form effective regions, we establish a relationship structure across the points that adapt to the distribution of features, considering both local and regional contexts. Specifically, for adaptive sampling points P, where each point p is associated with a feature vector f_p from feature map F, we establish relationships between these points that reflect both feature similarity and spatial proximity across varying object densities and scale. To capture these relationships, we first project point features f_p into a relation-sensitive space [see (7)]. The feature similarity between any two points p and q is then computed by the dot product $(\cdot)$, with higher values indicating stronger feature-based relationships [see (8)]. Next, to incorporate spatial proximity, we calculate the spatial displacement vector $\Delta_{pq} = p - q$. This vector represents the relative spatial location between the two points, which is then passed through a spatial weighting function $sp(\Delta_{pq})$ to generate a weighted representation that emphasizes relevant spatial relationships [see (9)]. To create the final relationships $m(p, q)$, we modulate the feature similarity score with this spatial proximity weight, emphasizing relationships between

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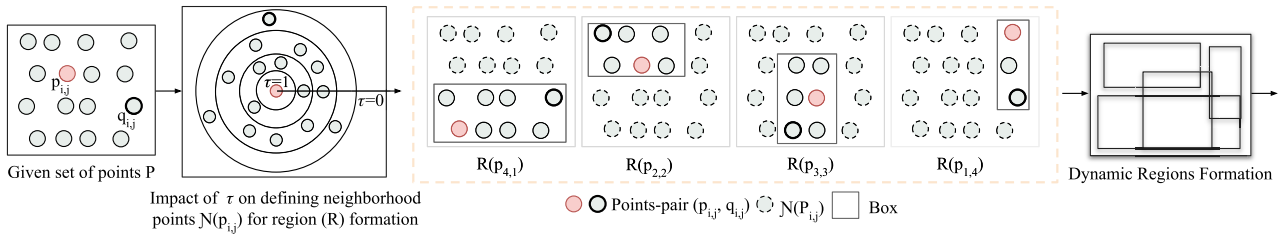


Fig. 5. Thresholding mechanism and neighborhood formation in dynamic regions formation. This illustration shows how the threshold τ controls the proximity of neighbors. As τ increases, only points within a certain distance are considered neighbors based on their similarity. Different regions $R_{(p)}$ are formed by varying the neighborhood criteria, with boxes representing the dynamically formed regions. These regions help to focus attention on relevant areas while ignoring irrelevant points, ensuring efficient detection.

spatially close and similar-feature points, while deemphasizing distant or dissimilar ones [see (10)]

$$h_p = \sigma\big(\text{linear}\big(f_p\big)\big), \quad h_p \in \mathbb{R}^H \tag{7}$$

$$fs(p,q) = \text{softmax}\left(\frac{h_p \cdot h_q^\top}{\sqrt{H}}\right) \tag{8}$$

$$sp\big(\Delta_{pq}\big) = \text{FFN}\big(\Delta_{pq}\big) \tag{9}$$

$$m(p,q) = fs(p,q) \cdot sp\big(\Delta_{pq}\big) \tag{10}$$

where $(H)^{1/2} = 32$ and σ denotes the GeLU.

The computed relationship scores $m(p,q)$ are then used to define point neighborhoods through a dynamic threshold mechanism τ [see (11)]. It selects, for each point p, a subset of points q that exhibit a strong feature and spatial relationship. A larger τ results in smaller, more selective neighborhoods $\mathcal{N}(p)$, whereas a smaller τ allows the inclusion of more points, leading to larger, less selective neighborhoods, as illustrated in Fig. 5

$$\mathcal{N}(p) = \left\{ q \in P \mid m(p,q) > \tau \cdot \max_{q \in P} m(p,q) \right\}. \tag{11}$$

Each point p together with its discrete neighborhood $\mathcal{N}(p)$, forms a local cluster that can be mapped to the minimal rectangular box encompassing all points in that neighborhood. This mapping defines the region $\{R_p\}_{p \in P}$, which ensures spatial continuity while adapting to the local point distribution. Consequently, the resulting adaptive region is customized to the underlying data and the relationships among the adaptive points [see (12)]. The structured nature of R_p provides significantly better starting conditions compared to purely random initialization in DETR, effectively addressing a fundamental challenge in its convergence. The use of spatial priors in DART helps mitigate, rather than amplify, errors caused by random initialization during early training iterations. R_p serves as spatial anchors, allowing object queries to focus on specific regions of the image from the start of training. This structured foundation facilitates more efficient matching between object queries and target objects, reducing the need for extensive decoder refinement iterations and accelerating training convergence by guiding attention effectively

$$R_p = \{(x,y) \mid (x,y) \in \text{Box}(p \cup \mathcal{N}(p))\} \tag{12}$$

where Box$(\cdot)$ computes the minimal rectangular bounds containing all points with (x,y) coordinates.

5) Attention Computation: In contrast to prior methods [46], [47], which processes all spatial locations uniformly, we map each adaptive region R_p onto the original feature map X [see (13)] to capture the most semantically relevant region. This region-based approach prioritizes spatially continuous regions, leading to a more efficient and accurate feature aggregation

$$X_p = \big\{ X_{x,y} \mid (x,y) \in R_p \big\}. \tag{13}$$

These regional features X_p, then undergo a self-attention mechanism to capture relationships between the features within each region, refining the regional representation. Specifically, regional features are projected into the query, key, and value spaces [see (14)], and the attention for each region R_p is computed as shown in (15). In addition, we introduce a region-specific bias term B_p to prioritize spatially relevant interactions. This term encodes relative positional relationships within regions, acting as a lightweight spatial before guiding attention toward geometrically coherent features

$$Q_p = X_p W_q, \quad K_p = X_p W_k, \quad V_p = X_p W_v \tag{14}$$

$$\text{ARA}\big(X_p\big) = \text{softmax}\left(\frac{Q_p K_p^T}{\sqrt{d_k}} + B_p\right) V_p \tag{15}$$

where $W_q, W_k, W_v \in \mathbb{R}^{C \times d}$ are learnable projection matrices. The matrices $Q_p, K_p, V_p \in \mathbb{R}^{w_1 w_2 \times d}$ represent the query, key, and value for point p, respectively. $(d_k)^{1/2} = 512$, $B_p \in \mathbb{R}^{w_1^2 w_2^2}$, and $w_1 \times w_2$ indicates the dimensions of the region R_p.

B. Region-Aware Decoder

While DETR [13] and its variants [18], [38], [40] demonstrate potential in object detection, they suffer from attention redundancy, since the uniform attention mechanism compels queries to evaluate all spatial positions indiscriminately. This leads to computational inefficiencies and missed detections, particularly when dealing with tiny objects across varying scene complexities. To address these limitations, we designed RAD, which builds upon the DETR decoder architecture by replacing the standard cross-attention with MRA, as demonstrated in Fig. 3 (left). Unlike previous DETR-like methods that take the entire flattened feature map as input, RAD utilizes only masked regions from our encoder to dynamically focus on semantically significant areas. This approach prioritizes distinct and spatially coherent features, enhancing the model's

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ability to resolve ambiguities in dense or visually homogeneous environments. Given a set of queries $Q_{\text{DETR}} = \{q_i\}_{i \in N}$ and regions $\{R_p\}_{p \in P}$, we extract the masked region features $\{F_p\}_{p \in P}$ from I_{enc} using regions R_p. This extraction utilizes a binary mask [see (16)] on the encoder output features I_{enc}, as shown in Fig. 6(a). This process preserves the content within the adaptive regions R_p while suppressing other spatial locations, defined as F_p [see (17)]. Whereas I_{enc} extracts multiple dynamic regions, we utilize 125 most informative regions based on density, as input to our decoder

$$\mathcal{M} = \bigcup_{p \in P} \mathbb{I}\big[(x, y) \in R_p\big] \tag{16}$$

$$F_p = I_{\text{enc}} \odot \mathcal{M} \tag{17}$$

where N represents the total number of queries, set to $N = 300$. M denotes the binary mask, $\mathbb{I}$ is the indicator function for spatial coordinates, $M \in \{0, 1\}^{H \times W}$, and $\odot$ indicates the Hadamard product.

Once the masked regions are formed, the density-center point c_p is chosen to represent the most densely populated location within the region, reflecting the area with the highest concentration of objects and making it semantically meaningful. The point c_p serves as an effective spatial anchor for explicitly encoding the relationship between the queries q_i and the masked region F_p. Based on this, a region-query bias $M_{q,p}$ is introduced as depicted in Fig. 6(a). This bias enhances alignment with semantically rich areas, ensuring that attention is directed toward regions with a higher probability of containing relevant objects [see (18)]. $M_{q,p}$ guides queries with similar spatial priors (e.g., detecting adjacent objects) toward their expected locations, minimizing redundant attention on irrelevant areas

$$M_{q,p} = \text{linear}\big[\big(Q_{\text{DETR}} \| c_p\big)\big]. \tag{18}$$

Building on these masked regions F_p, we compute the attention weights using MRA. This mechanism accounts for the spatial relationships between queries and the selected regions, ensuring that each query attends to the most relevant areas corresponding to its assigned object [see (20)]. Specifically, for each query q_i, we calculate the attention weight, which quantifies how much q_i attends to the associated masked region feature F_p. By focusing queries on distinct regions, RAD effectively mitigates redundant attention that may arise when multiple queries attend to irrelevant areas. This strategy enhances each query's ability to capture relevant features based on the spatial coherence and semantic relevance of the region it attends to

$$Q = Q_{\text{DETR}} W_q, \quad \text{K}_p = F_p W_k, \quad V_p = F_p W_v \tag{19}$$

$$\text{MRA}\big(\text{Q}_{\text{DETR}}, \text{F}_\text{p}\big) = \sum_{p \in P} \text{softmax}\left(\frac{Q \text{K}_p^T}{\sqrt{d_k}} + M_{q,p}\right) V_p \tag{20}$$

where W_q, W_k, and W_v are the learned weight matrices.

Following DETR, these representations are then fed into the predictions feed-forward networks (FFNs), which predict bounding box coordinates and object categories. Importantly, RAD utilizes only the top-k attended masked regions for each query, ensuring that predictions are based on the most salient areas. When RAD processes these selected regions, it focuses on their most relevant components through masked cross-attention mechanisms. Each region is treated independently, allowing the decoder to predict bounding boxes and class labels for the objects within the specific region. If a region contains no objects, the decoder outputs "no object" predictions for those areas. By integrating spatial context with the feature relationships learned by the encoder, RAD efficiently localizes and identifies objects while maintaining computational efficiency, ultimately reducing the complexity of processing the entire image.

C. Query Diversity Loss

During training, DETR assigns a fixed set of object queries to ground-truth objects using a Hungarian matching algorithm. This method minimizes a total cost that includes both classification loss (LCls) and bounding box regression loss (L1 and LGIoU). While the Hungarian loss ensures that each query effectively identifies objects, it does not inherently prevent multiple queries from focusing on the same region, as depicted in Fig. 6(b), (left). In complex scenes, particularly when objects are tiny or densely packed, several queries may attend to identical or overlapping spatial areas. This redundancy leads to two main issues: 1) a reduction in the overall discriminative power of the feature representation, as overlapping queries fail to capture complementary or diverse object characteristics and 2) an increased likelihood of missing subtle details or distinguishing between closely situated objects, which is crucial for accurately localizing tiny or partially occluded objects. The Hungarian loss is expressed as follows:

$$\text{L}_{\text{Hungarian}} = \text{L}_{\text{Cls}} + \lambda \text{L}_1 + \lambda \text{L}_{\text{GIoU}} \tag{21}$$

where $\lambda_{\text{L1}} = 5$ and $\lambda_{\text{GIoU}} = 2$.

To address this issue, we propose the query diversity loss, which penalizes overlapping attention distributions among different queries by explicitly measuring the similarity between the feature vectors produced by these queries when they attend to the same region, as illustrated in Fig. 6(b). This approach employs the cosine similarity metric, which quantifies the alignment between two vectors. When queries focus on the same region similarly, their cosine similarity value increases. The diversity loss penalizes these high similarity values, encouraging the model to adjust the queries so that each one extracts distinct and complementary information. Minimizing the query diversity loss encourages the model to diversify its attention across the scene. As a result, each query is guided to capture unique spatial details instead of redundantly attending to the same features. The query diversity loss is formulated as follows:

$$\text{L}_{\text{Diversity}} = \sum_{q_1, q_2 \in Q \,|\, q_1 \neq q_2} \sum_{p \in P} \cos\big(f_{q_1,p}, f_{q_2,p}\big) \tag{22}$$

where $f_{q,p}$ represents the feature vector for query q attending to region R_p, and the cosine similarity $\cos(\cdot)$ measures the similarity between the attended feature vectors of two different queries q_1 and q_2.

This combined loss function ensures that the model not only makes accurate object predictions (via the Hungarian loss) but

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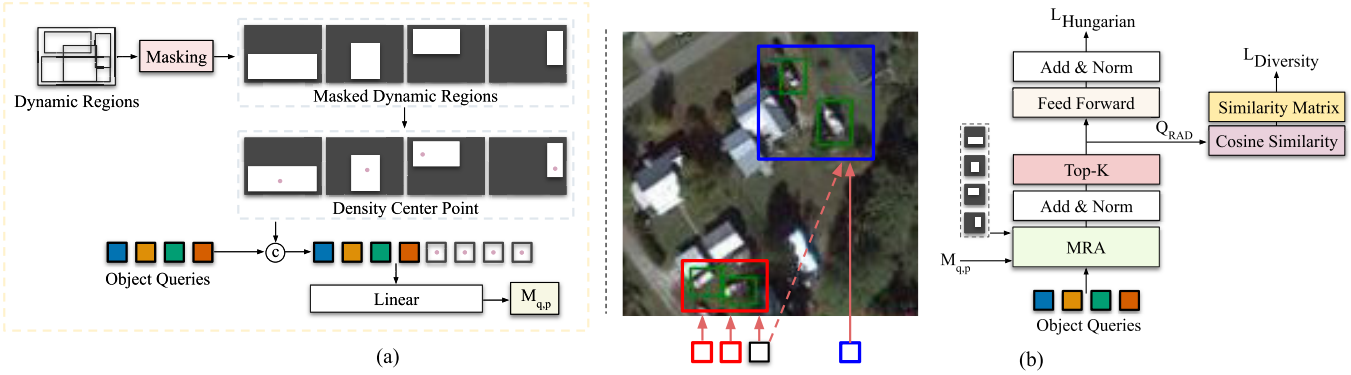


Fig. 6. Illustration of the components within RAD. (a) Dynamic regions formed by the DAT are isolated through masking to highlight relevant areas, which are then associated with a density center point. This point is used to define object queries, which are processed through a linear layer to generate the region-query bias, $M_{q,p}$, guiding attention to the most important areas. (b) Diversity loss mechanism ensures that object queries focus on distinct regions (dotted soft red) despite overlapping regions (solid soft red) by calculating the cosine similarity in the similarity matrix and applying a diversity loss to penalize overlapping attention. The Hungarian algorithm is then used to match queries optimally, improving the model's ability to differentiate between objects and enhancing its overall efficiency in detecting diverse objects.

also avoids redundancy in the attention mechanism by focusing on diverse regions (via the query diversity loss). Consequently, the model becomes more adept and efficient at distinguishing between overlapping or similar objects, which is effective in both dense and sparse environments with varying object sizes. The total loss function for RAD is defined as

$$L_{\text{RAD}} = \text{L}_{\text{Hungarian}} + \lambda \text{L}_{\text{Diversity}} \tag{23}$$

where $\lambda = 0.45$ is a hyperparameter that controls the relative importance of the query diversity loss. Moreover, we incorporate prediction FFNs and total loss after each decoder layer, with all prediction FFNs sharing the same parameters.

IV. Experiments

A. Datasets

1) AI-TOD: This is a large-scale publicly available dataset [1] for advancing tiny-object detection in optical remote sensing imagery. It presents complex challenges through a variety of scenes including parking lots, residential areas, marine environments, and dense fog. AI-TOD comprises 28 036 aerial images, each with a size of 800 × 800 pixels, containing over 700 000 instances across eight diverse object categories. Notably, the average object size is only 12.8 pixels, with 86% of objects being smaller than 16 pixels. Object sizes are further categorized into very tiny (vt) for objects within [2, 8] pixels, tiny (t) for [8, 16] pixels, small (s) for [16, 32] pixels, and medium (m) for [32, 64] pixels. The dataset is divided into training and testing sets, with each containing 14 018 images.

2) DOTA-v2.0: This dataset [49] serves as a large-scale benchmark for object detection in aerial images. It contains real-world scenarios across diverse landscapes and terrains, reflecting the complexities of object detection in challenging environments. The dataset includes 11 268 images, and 1 793 658 instances, ranging from 800 × 800 to 4000 × 4000 pixels, with a significant number of tiny objects (smaller than 10 pixels) across 18 categories. Following [50], we use the training set for training and the validation set for testing.

TABLE I
Impact of Adaptive Sampling Point Generation Used in DAT During Stage 1

Local Feature	Spatial Encoding	Adaptive Sampling	AP	AP_{50}	AP_{vt}	AP_t
×	✓	✓	24.1	55.8	12.2	11.4
✓	✓	×	25.5	57.1	12.4	12.9
✓	×	✓	20.2	50.4	08.4	17.4
✓	✓	✓	30.6	65.4	17.7	33.3

3) LEVIR-Ship: This dataset [2] presents a significant challenge for tiny-ship detection due to its diverse and complex background conditions. This includes variations in cloud cover, land presence, illumination, and sea area characteristics. Comprising 3896 optical images of 512 × 512 pixels, it contains 3000 instances of ships, with an average object size of only 10 pixels. Notably, approximately 90% of the ships are smaller than 20 pixels, and the largest ship is no more than 32 pixels. The dataset is divided into training, validation, and testing sets, consisting of 2320, 788, and 788 images, respectively.

4) DIOR: To validate the generalization of the proposed method for handling large-scale multiclass objects, we additionally introduce the DIOR [51] dataset. It features a wide range of object sizes, diverse image variations, high interclass similarity, and intraclass diversity. It contains 23 463 images and 192 472 object instances, with each image sized at 800 × 800 pixels and an average object size of 65 pixels. The dataset includes 20 distinct object categories, including airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), golf course (GC), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill (WM). The dataset is divided into training and testing sets, with each containing 11 000 images.

B. Implementation Details and Evaluation Metrics

The experiments were conducted on a single NVIDIA RTX 3090 GPU, utilizing the PyTorch framework and the MMDetection toolbox. For the baseline, we modify DETR

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TABLE II
IMPACT OF LEARNING POINT RELATIONS USED IN DAT DURING STAGE 2 TO DEFINE POINT NEIGHBORHOODS FOR REGION FORMATION

Feature Similarity	Spatial Proximity	AP	AP_{50}	AP_{vt}	AP_t
×	×	24.5	57.3	10.6	25.1
✓	×	28.3	61.7	14.3	30.7
✓	✓	30.6	65.4	17.7	33.3

TABLE III
IMPACT OF DYNAMIC THRESHOLD MECHANISM USED IN DAT DURING STAGE 2 TO MODULATE NEIGHBORHOODS SIZE FOR REGION FORMATION

Temperature Value	AP	AP_{50}	AP_{vt}	AP_t
Set to Low with $\tau = 0.01$	24.5	57.0	10.0	25.0
Set to Medium with $\tau = 0.10$	29.0	62.5	15.0	30.5
Set to High with $\tau = 1$	25.0	57.5	10.5	25.5
Set to Dynamic	30.6	65.4	17.7	33.3

TABLE IV
IMPACT OF REGION-SPECIFIC BIAS ON REGION-ADAPTIVE ATTENTION USED IN DAT DURING STAGE 3

Region Specific Bias	AP	AP_{50}	AP_{vt}	AP_t
×	27.5	60.8	13.4	28.2
✓	30.6	65.4	17.7	33.3

[13] with Swin-T [47] as the backbone. The input image sizes for the datasets were: AI-TOD (800 × 800), LEVIR-Ship (512 × 512), DOTA-v2.0 (800 × 800), and DIOR (800 × 800). For optimization, we employed the stochastic gradient descent (SGD) optimizer with a momentum of 0.9, an initial learning rate of 0.005, and a weight decay of 0.0001. The minimum training batch size was set to 8. The training epochs were set to 150 for both the AI-TOD and DOTA-v2.0 datasets, 120 for LEVIR-Ship, and 200 for the DIOR dataset. We adhered to the COCO evaluation metrics, where AP denotes the average precision. Specifically, AP_{50} indicates that the IoU threshold for defining a true positive (TP) is 0.5, while AP_{75} sets the threshold at 0.75. AP represents the average value calculated from AP_{50} to AP_{95}, with an IoU interval of 0.05. For AI-TOD, the metrics include AP, AP_{50}, AP_{75}, AP_{vt}, AP_t, AP_s, and AP_m. For LEVIR-Ship, the metric used is AP_{50}. For DOTA-v2.0 and DIOR datasets, the metric includes AP, AP_{50}, AP_{vt}, and AP_t. Note that for DIOR reported mAP is equivalent to AP_{50}. To visualize the detection bounding boxes, we categorize them as TPs, false negatives (FNs), and false positives (FPs) based on a confidence threshold of 0.3 [52]. Additionally, to evaluate the processing efficiency of the object detectors, we introduce metrics such as the number of parameters (Params), floating-point operations (FLOPs), and frames per second (FPS).

C. Ablation Study

All model variants used in Tables I–VIII are trained with our default settings, which include passing regions as input from DAT to the RAD and employing the L_{RAD} loss function unless specific settings are mentioned in the corresponding table for that version. All ablation studies are conducted on the AI-TOD test set.

1) Impact of Adaptive Sampling Point Generation: Table I analyzes the contributions of local feature extraction, adaptive sampling, and spatial encoding in generating sampling points within DAT. Removing local feature transformations results in a 6.5% AP drop, highlighting their role in detecting tiny objects. Substituting adaptive sampling with uniform sampling leads to a 5.1% decrease, emphasizing the importance of directing attention to relevant regions. Omitting spatial encoding causes a significant decline in AP to 20.2% and AP_{vt} to 08.4%, underscoring the necessity of global spatial context. Overall, the best performance (AP: 30.6%, AP_{vt}: 17.7%) is achieved when all components are active, demonstrating their collective importance.

2) Learning Point Relations for Region Formation: Table II examines the impact of learned point relationships in DAT during stage 2, focusing on feature similarity and spatial proximity for defining point neighborhoods. Replacing learned relationships with k-nearest neighbors ($k = 6$) significantly reduced performance to an AP of 24.5% and AP_{vt} of 10.6%, highlighting the limitations of fixed heuristics. In contrast, earned point relationships, while focusing solely on feature similarity improved AP and AP_{vt} by 3.8% and 3.6%, respectively. However, excluding spatial proximity decreased performance by 2.3% in AP and 3.4% in AP_{vt}. These results emphasize the importance of spatial proximity for maintaining geometrically coherent connections, with the full model achieving the best performance through dynamic modeling of object topology.

3) Impact of τ on Neighborhoods Size Selection: Table III examines how temperature τ influences neighborhood size during region formation in stage 2. Low temperatures result in larger neighborhoods that dilute discriminative features, while medium temperatures achieve an optimal balance of point inclusion and filtering. In contrast, high temperatures create overly selective neighborhoods that restrict spatial context. The dynamic approach enhances performance by adapting neighborhood sizes to scene complexity, allowing smaller regions to prevent feature contamination in dense areas and larger regions to maintain coherence in sparse environments.

4) Impact of Region-Specific Bias: Table IV evaluates the impact of the region-specific bias (B_p) term on region-adaptive attention in stage 3. Its removal results in a 3.1% drop in AP and significant declines in all metrics, impairing the model's ability to prioritize spatial interactions and negatively affecting small-object detection and localization. This underscores the importance of even minor spatial priors for stabilizing feature alignment and enhancing robustness to scale variations.

5) Comparison of Decoder Variants: Table V compares two design variants of the DETR decoder with our proposed model. Both variants utilize flattened feature maps from our DAT, but they differ in their use of spatial positional encoding (P.E.). The variant without this encoding demonstrates the lowest performance (25.7% AP) because rigid spatial aggregation limits feature extraction, particularly for tiny objects (11.8% AP_{vt}). While adding spatial P.E. results in a slight performance improvement (AP_{50}: +1.0%, AP_{vt}: +0.7%), it still retains

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TABLE V
COMPARISON OF DETR DECODER VARIANTS WITH THE PROPOSED METHOD. P.E. DENOTES POSITIONAL ENCODING

Configurations	AP	AP_{50}	AP_{vt}	AP_t
DETR-Decoder	25.7	57.5	11.8	26.3
DETR-Decoder+Spatial P.E.	26.0	58.5	12.5	26.9
Full Model	30.6	65.4	17.7	33.3

TABLE VI
IMPACT OF REGION QUERY-BIAS ON MASKED REGION CROSS-ATTENTION USED IN RAD

Region-Query Bias	AP	AP_{50}	AP_{vt}	AP_t
×	27.8	60.4	14.3	29.7
✓	30.6	65.4	17.7	33.3

TABLE VII
IMPACT OF QUERY DIVERSITY LOSS

$L_{Hungarian}$	$L_{Diversity}$	AP	AP_{50}	AP_{vt}	AP_t
✓	×	27.4	60.6	13.6	28.4
✓	✓	30.6	65.4	17.7	33.3

DETR's global attention mechanism, which disperses focus across irrelevant regions and constrains overall gains. In contrast, our model achieves the highest performance (30.6% AP) by employing adaptive regions, masked region-specific cross-attention, and region-query bias, significantly enhancing the detection of very small objects (17% AP_{vt}).

6) Impact of Region-Query Bias: Table VI assesses the effect of region-query bias ($M_{q,p}$) on masked region cross-attention within RAD. Removing this bias results in a performance drop (AP: 27.8% versus 30.6%), as it impairs the model's ability to adapt queries to localized contexts, losing guidance for feature prioritization. In contrast, the model with region-query bias shows a significant performance increase (AP +2.8%), enhancing query-region alignment, refining feature interaction, and improving the detection of objects in specific areas.

7) Impact of Query Diversity Loss: Table VII examines the effect of query diversity loss ($L_{Diversity}$) on performance. Omitting this loss leads to a decrease in average precision (AP) to 27.4%, underscoring its significance in preserving distinct queries and preventing feature overlap. In contrast, the full model that incorporates query diversity loss achieves an AP of 30.6%, ensuring comprehensive coverage of object features. This enhancement significantly improves the detection of very small objects (AP_{vt}: 17.7%, AP_t: 33.3%) while also reducing redundancy.

8) Impact of Proposed Encoder–Decoder Architecture: Fig. 7 and Table VIII demonstrate that the proposed DAT and RAD improve performance significantly while lowering complexity compared to the baseline model, which uses the Swin-T as the encoder and DETR decoder at default settings. The baseline model achieves moderate AP (18.4%) and AP_{50} (41.4%) scores with 44 million parameters and 254 GFLOPs, struggling with tiny-object detection. For the DAT + DETR decoder, we use flattened feature maps from DAT instead of masked regions and include spatial P.E. in the decoder. In the Swin-T + RAD version, we utilize features from various Swin layers along with P.E. Compared to Swin-T, DAT enhances AP and AP_{50} by 7.4% and 20.3%, respectively, while reducing parameters and FLOPs by 36.4% and 45%, and improving FPS by 11%. Similarly, RAD outperforms the DETR decoder, increasing AP and AP_{50} by 8% and 19.1%, while decreasing parameters and FLOPs by 18.2% and 34.6%, with a slight increase of 2 FPS. Both DAT and RAD significantly enhance tiny and very tiny APs, improving the baseline model's ability to detect very small objects. The full model DART achieves the highest performance, with an AP of 30.6% and an AP_{50} of 65.4%. It shows significant improvements in AP_{vt} (17.7%) and AP_t (33.3%) while reducing computational requirements to 13 million parameters and 68 GFLOPs, and increasing FPS to 45. This illustrates that region-aware mechanisms effectively enhance both accuracy and efficiency for tiny-object detection.

TABLE VIII
IMPACT OF THE PROPOSED ENCODER–DECODER ARCHITECTURE

Encoder	Decoder	AP	AP_{50}	AP_{vt}	AP_t	Params	FLOPs	FPS
×	×	18.4	41.4	5.3	13.2	44 M	254 G	27
✓	×	25.8	61.7	13.2	26.9	28 M	140 G	30
×	✓	26.4	60.5	12.5	27.8	36 M	166 G	25
✓	✓	30.6	65.4	17.7	33.3	13 M	68 G	45

D. Comparison With the State-of-the-Arts

1) Results on AI-TOD: Fig. 8 and Table IX present the performance comparison on the AI-TOD test set. The results for the competing state-of-the-art methods have been sourced from the CAF^{2}ENet [61], MENet [7], SWIN-TOD [60], and related articles. These methods include representative generic object detectors and tiny-object detectors specifically designed for remote sensing imagery. Among generic detectors, end-to-end Transformer-based models demonstrate superior overall performance, with DINO-deformable-DETR [36] leading at an AP of 23.8%. However, it still lags behind the performance of the DETR-based tiny-object detector, DQ-DETR [18]. In tiny-object detection, SAFF-SSD [32], despite employing parallel local and global self-attention, achieves an AP of only 21% along with a low AP_{vt} of 7.0%. STDet [5], which utilizes global deformable attention and local attention, improves AP to 25.5%, but experiences a decline in AP_m from 38.8% to 35.1%. Multiscale feature fusion methods, including MAV [34], FSANet [57], MENet [7], ORFENet [24], and FFCA-YOLO [58], further improve detection networks but show limited improvements in AP ranging (17.2→23.9) and AP_{vt} (08.9→09.4). Among these, FFCA-YOLO achieves a notable AP of 27.7% and an AP_{vt} of 12.6%, yet still lags behind STDet in AP_t. In contrast, DNTR [8] utilizes self-attention in the detection head to enhance the representation of very small objects and achieves an AP_t of 26.4%. However, its performance is constrained to 11.9 FPS due to increased computational overhead. Label assignment strategy-based models, including NWD-RKA [52] to SimD [29] enhance detection performance by assigning more positive samples to tiny objects. Specifically, NWD-RKA with the

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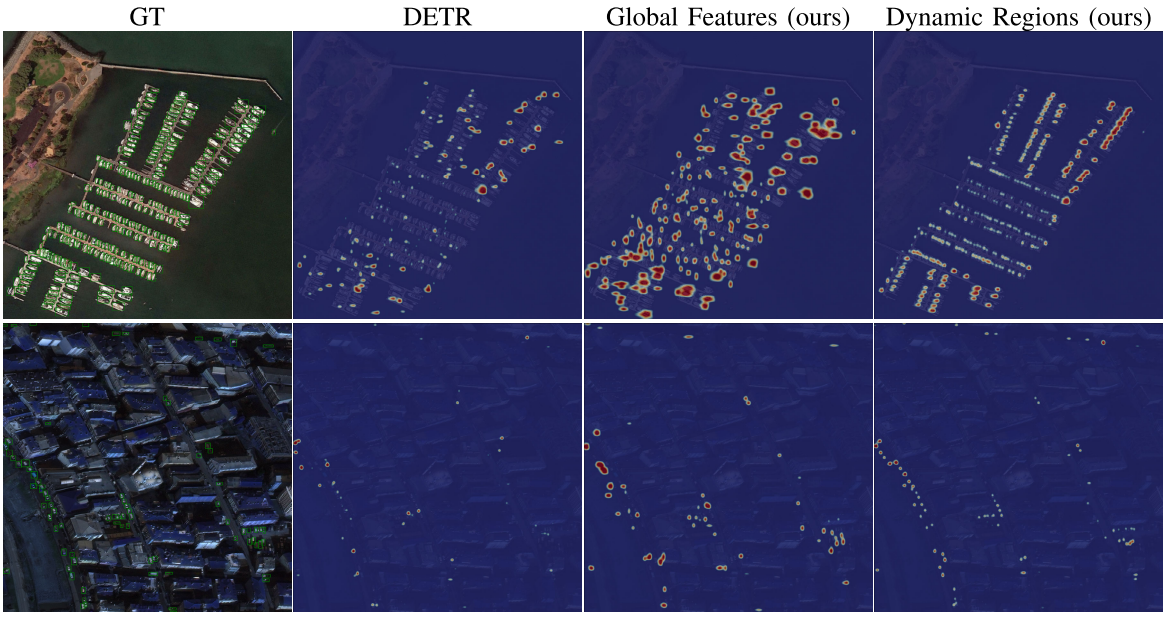


Fig. 7. Visualization of features passed to the decoder for different methods. The first column (GT) shows the ground-truth annotations. The second column presents the DETR model's output, where the attention map highlights regions of interest. The third and fourth columns display the results of the proposed methods: global features maps and dynamic regions, respectively. These attention maps illustrate the model's focus on different areas, with dynamic regions providing more refined localization and better spatial attention compared to the DETR and flattened regions approaches.

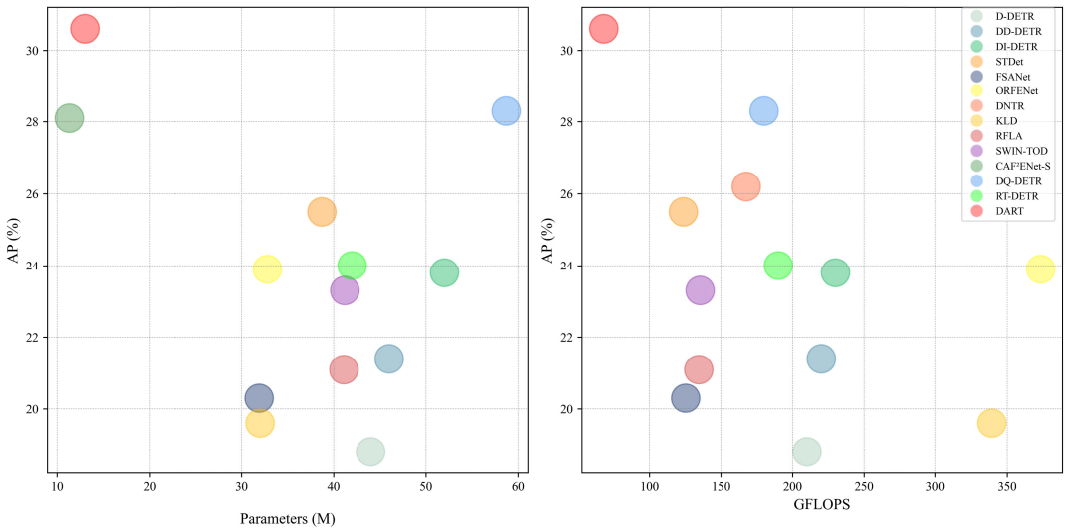


Fig. 8. AP versus model parameters (M) and GFLOPs tradeoff on the AI-TOD dataset. DART, highlighted in red, outperforms state-of-the-art models while utilizing resources efficiently. D-DETR refers to deformable-DETR, DD-DETR to DAB-deformable-DETR, and DI-DETR to DINO-deformable-DETR.

Swin-T achieves an AP of 22.2% and TTFNet-GA [28] with MiT-B1 boosts it to 24.2%, but both have modest AP_{vt} gains due to sensitivity to localization errors. SWIN-TOD [60] attains 25.9% AP, yet is biased toward larger objects, improving AP_{vt} by only 0.4%. SimD achieves 13.4% AP_{vt} due to its location-shape similarity metric, but its reliance on NMS results in tradeoffs, with AP_{50} at 55.9% and AP_{75} at 21.2%. Conversely, BCCResNet [10], despite training a separate network for each category based on dataset difficulty levels, achieves performance similar to SimD.

CAF^{2}ENet [61] improves AP_{vt} to 12.8% and AP_s to 36.7%, but its use of cross-attention-based upsampling increases computational overhead, resulting in an overall AP of 30.2% at 31.6 FPS. DQ-DETR [18], with a deformable encoder–decoder, achieves an AP of 28.3% and an AP_{vt} of 13.9%, making it the second-best after CAF^{2}ENet. However, its two-stage training and reliance on accurate query counting add complexity, making it vulnerable to misclassification in dense or sparse scenes. In contrast, our DART model optimizes attention allocation for varying object densities in dynamic, content-aware spatial regions, minimizing computational burden while enhancing detection capabilities for different object sizes in complex environments. With 13 M parameters, 68 GFLOPs, and 45 FPS, DART achieves the highest AP (30.6%), AP_{50} (65.4%), and AP_{75} (28.4%). It outperforms second-best methods with improvements of 3.8% in AP_{vt}, 3.0% in AP_t, and

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TABLE IX

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE AI-TOD TEST SET. RED DENOTES THE BEST RESULTS, AND BLUE SECOND-BEST. † INDICATES METHODS EMPLOYING THE SELF-ATTENTION MECHANISM IN THE DETECTION FRAMEWORK (I.E., BACKBONE, FPN, AND DETECTION HEAD), AND ‡ INDICATES DETR-LIKE METHODS

Method	Backbone	Venue	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
Generic object detector									
Faster R-CNN [3]	ResNet-50	NIPS15	11.4	27.0	08.0	00.0	08.3	23.1	24.5
RetinaNet [53]	ResNet-50	ICCV17	04.7	13.6	02.1	02.0	05.4	06.3	07.6
Cascaded R-CNN [54]	ResNet-50	CVPR18	11.8	25.6	09.5	00.1	07.9	23.0	34.8
RepPoints [55]	ResNet-50	ICCV19	09.2	23.6	05.3	02.5	09.2	12.9	14.4
†FCOS [4]	Swin-T	ICCV19	16.2	48.9	11.5	08.0	19.1	26.1	28.2
†TTFNet [56]	MiT-B1	AAAI20	17.6	43.5	12.6	04.3	15.1	15.8	32.3
‡ DETR [13]	Swin-T	ECCV20	18.4	41.4	16.2	05.3	13.2	23.7	31.5
‡ Deformable-DETR [16]	Swin-T	ICLR21	18.8	47.4	10.2	09.1	18.7	22.9	29.6
‡ DAB-Deformable-DETR [38]	Swin-T	ICLR22	21.4	49.4	14.5	11.0	20.5	27.2	36.4
‡ DINO-Deformable-DETR [36]	Swin-T	ICLR23	23.8	57.3	16.1	11.3	24.2	30.5	38.3
Tiny object detector									
FSANet [57]	ResNet-50	TGRS22	20.3	48.1	14.0	06.3	19.0	26.8	36.7
† NWD-RKA [52]	Swin-T	ISPRS22	22.2	55.2	13.4	08.6	22.1	27.2	35.3
RFLA [50]	ResNet-50	ECCV22	21.1	51.6	13.1	09.5	21.2	26.1	31.5
† SAFF-SSD [32]	2LTransformer	REMS23	21.1	49.9	14.9	07.0	20.8	30.1	38.8
† MAV [34]	Swin-T	MAV23	17.2	47.7	08.0	08.9	18.1	21.2	28.4
ADAS-GPM [27]	ResNet-50	NEUC23	24.1	55.2	17.3	08.9	24.0	29.3	37.2
† MENet [7]	Swin-T	TGRS24	23.2	56.2	15.0	09.7	23.9	25.3	34.4
ORFENet [24]	ResNet-50	TGRS24	23.9	54.5	18.0	09.4	23.8	26.9	34.4
FFCA-YOLO [58]	CSPDarkNet-53	TGRS24	27.7	61.7	-	12.6	24.9	31.2	-
† DNTR [8]	ResNet-50	TGRS24	26.2	56.7	20.2	12.8	26.4	31.0	37.0
DILA [59]	ResNet-50	ASOC24	23.0	52.6	16.6	07.9	23.5	28.8	34.6
† TTFNet-GA [28]	MiT-B1	TGRS24	24.2	53.9	19.4	10.4	25.2	24.5	34.1
SWIN-TOD [60]	ResNet-50	TGRS24	23.3	55.5	16.6	11.9	22.3	27.3	32.5
SimD [29]	ResNet-50	IROS24	26.6	55.9	21.2	13.4	27.5	30.9	37.8
BCCResNet [10]	ResNet-50	NCAA24	23.2	55.9	15.1	11.3	25.4	28.8	38.6
† CAF^2ENet-M [61]	CSPDarkNet-53	TGRS24	**30.2**	**63.7**	25.0	12.8	**30.3**	**36.7**	**41.9**
† STDet [5]	LGFormer	TGRS25	25.5	57.5	18.7	12.4	26.4	29.0	35.1
‡ DQ-DETR [18]	ResNet-50	ECCV25	28.3	60.4	24.6	**13.9**	28.7	31.4	38.6
DART	DAT		**30.6**	**65.4**	**28.4**	**17.7**	**33.3**	**38.0**	**42.4**

TABLE X

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V2.0 VAL SET. RED DENOTES THE BEST RESULTS, AND BLUE SECOND-BEST. ‡ INDICATES DETR-LIKE METHODS

Method	AP	AP_{50}	AP_{vt}	AP_t
Generic object detector				
Faster R-CNN [3]	35.6	59.5	00.0	07.1
Cascaded R-CNN [54]	37.0	59.5	00.0	05.9
FCOS [4]	31.8	55.4	00.3	04.0
‡DETR [13]	37.5	60.6	00.8	06.2
Tiny object detector				
ORFENet [24]	41.7	67.4	04.7	11.8
RFLA [50]	41.3	64.2	02.1	10.8
NWD-RKA [52]	41.9	66.3	01.9	10.6
‡DQ-DETR [18]	**42.3**	**69.5**	**05.6**	**13.1**
DART	**43.2**	**72.7**	**05.8**	**13.5**

TABLE XI

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE LEVIR-SHIP TEST SET. RED DENOTES THE BEST RESULTS, AND BLUE SECOND-BEST. † INDICATES METHODS EMPLOYING SELF-ATTENTION MECHANISM WITHIN BACKBONE, AND ‡ INDICATES DETR-LIKE METHODS

Method	AP_{50}	Method	AP_{50}
Generic object detector			
Faster R-CNN [3]	66.7	Cascaded R-CNN [54]	66.3
Mask R-CNN [62]	80.6	EfficientDet [63]	80.9
Yolov3 [64]	72.7	Yolov8 [65]	72.2
†FCOS [4]	77.2	‡DETR [13]	76.2
Tiny object detector			
†RTD-Net [66]	73.8	†SGMFNet [30]	74.1
DS-YOLOv5 [67]	75.4	DSFPAPNet [68]	82.6
ORFENet [24]	83.3	†DRENet [2]	82.4
RNet [69]	83.7	ADLA [70]	83.8
‡DQ-DETR [18]	**84.6**	DART	**84.9**

1.3% in AP_s. Fig. 9 illustrates visualization results on densely packed and sparsely distributed tiny objects, significantly demonstrating DART's performance in accurately identifying the variations.

2) Results on DOTA-v2.0: Table X compares performance on the DOTA-v2.0 validation set, which includes extremely small (less than 10 pixels) and larger instances (greater than 300 pixels). The results for these competing state-of-the-art approaches are reported from RFLA [50] and related publications. Among generic object detectors, DETR [13] outperforms two-stage cascade R-CNN [54] and achieves an AP_{vt} of 0.08%. In tiny-object detection, ORFENet [24] achieves an AP_{vt} of 04.7% and AP_t of 11.8%, surpassing RFLA [50] and NWD-RKA [52]. DQ-DETR [18] further improves AP to 42.3% and AP_{50} to 69.5%. In contrast, DART achieves the highest AP of 43.2%, with an AP_{50} of 72.7%, and improves AP_{vt} to 05.8% and AP_t to 13.5%. These results highlight DART's effectiveness in large-scale aerial image

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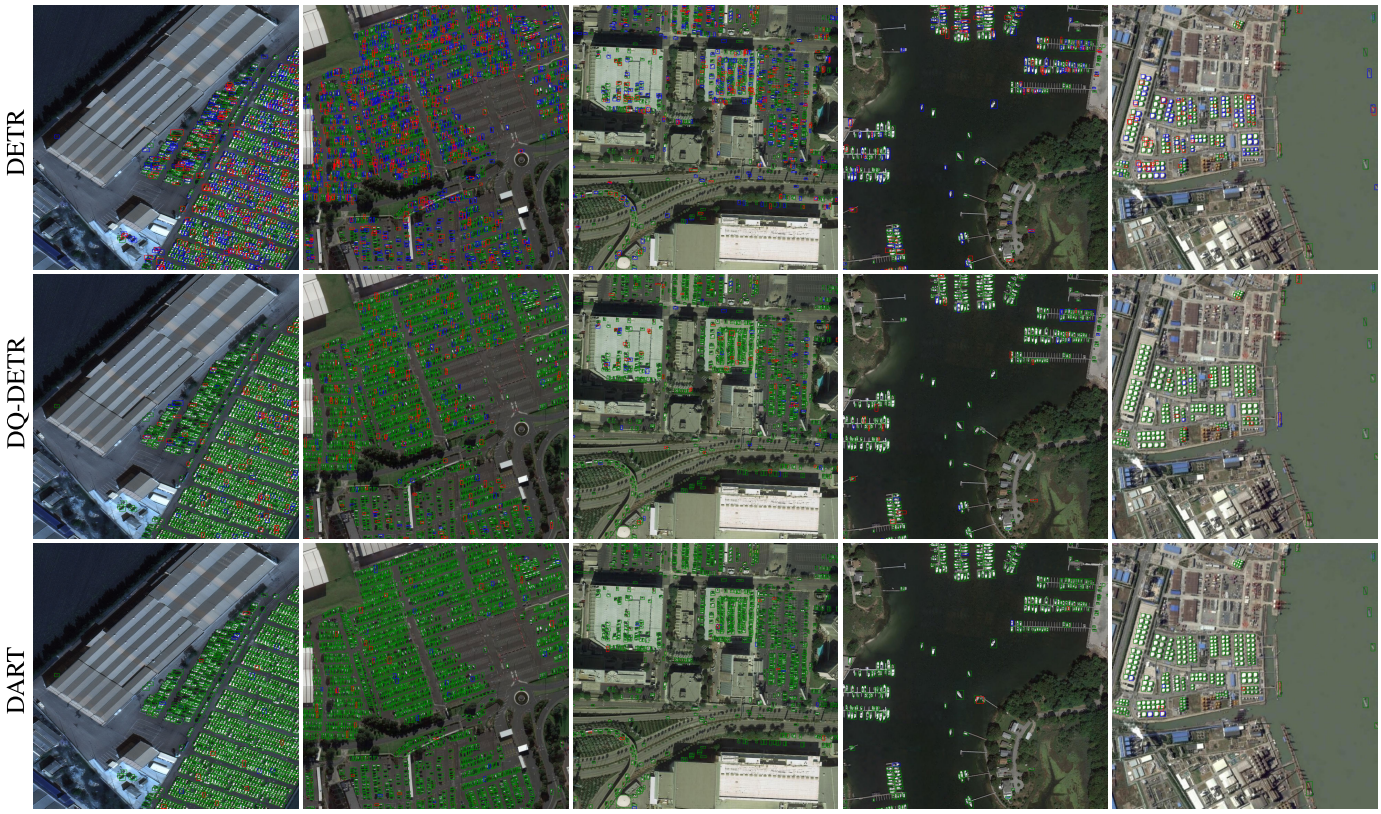


Fig. 9. Visualization results on the AI-TOD test dataset in densely packed settings. The green, blue, and red boxes represent TP, FP, and FN, respectively.



Fig. 10. Visualization results on the DOTA-v2.0 val dataset in densely packed maritime settings with larger objects across different platforms. The green, blue, and red boxes represent TP, FP, and FN, respectively.

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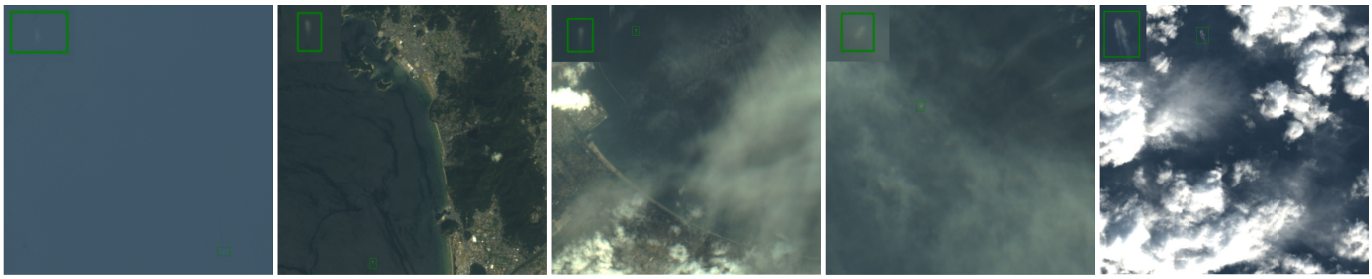


Fig. 11. Visualization results of DART for very tiny, sparse ships under challenging conditions on the LEVIR-Ship test dataset, with zoomed-in image slices in the top left corner for better observation.



Fig. 12. Visualization results of DART's generalization capability in handling various object sizes on the DIOR test dataset.

object detection, effectively addressing diverse landscapes, maritime environments, and terrains, as illustrated in Fig. 10.

3) Results on LEVIR-Ship: Table XI compares the performance of state-of-the-art tiny-object detectors designed for remote sensing imagery on the LEVIR-Ship test set. The results for these competing state-of-the-art approaches are reported from DRENet [2], ADLA [70], DS-YOLOv5 [67], and related publications. Our baseline DETR [13] shows comparable performance to CNN-based methods, highlighting its end-to-end detection capabilities. While RTD-Net [66], SGMFNet [30], and DS-YOLOv5 struggle with the complexities of satellite imagery, DRENet demonstrates a significant improvement in AP_{50}, rising from 75.4% to 82.4%. DQ-DETR [18] further enhances this performance by achieving an AP_{50} of 84.6%. However, DART achieves the highest AP_{50} of 84.9%, effectively detecting very small ships under varying cloud coverage and strong sea waves in diverse and complex background conditions, as shown in Fig. 11.

4) Results on DIOR: To further demonstrate the generalization capabilities of the proposed method, we performed experiments on the DIOR test set, analyzing results both class-wise (Table XII) and for tiny objects (Table XIII). The results for the competing state-of-the-art methods have been sourced from KLDet [11], CAF^{2}ENet [61], and related articles. Table XII shows class-wise results for objects ranging (0→800 pixels) in the DIOR dataset. Generic object detectors struggle to achieve satisfactory performance across different classes, exhibiting a wide range of mAPs (45.0→69.9). The results highlight that while our method is designed for tiny objects, it performs exceptionally well compared to object detectors designed for large-scale object detection, with mAPs ranging (71.1→76.2), and DETR-based models showing mAPs (66.8→72.0). Furthermore, when compared to tiny-object detection methods optimized for large-scale detection, which have mAPs ranging (71.8→75.4), our tiny-object detector achieves the highest mAP of 79.5%, demonstrating

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TABLE XII

CLASS-WISE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR TEST SET. RED DENOTES THE BEST RESULTS, AND BLUE SECOND-BEST. ‡ INDICATES DETR-LIKE METHODS

Method	Backbone	Venue	APL	APO	BF	BC	BR	CH	DAM	ESA	ETS	GC	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
Generic object detector																							
Faster R-CNN [3]	ResNet-50	NIPS15	50.3	62.6	66.0	80.9	28.8	68.2	47.3	58.5	48.1	60.4	67.0	43.9	46.9	58.5	52.4	42.4	79.5	48.0	34.8	65.4	55.5
RetinaNet [53]	ResNet-101	ICCV17	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
YOLOv3 [64]	DarkNet-53	CVPR18	72.2	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.5	31.1	61.1	44.9	49.7	87.4	70.6	68.7	87.3	29.4	48.3	78.7	57.1
‡ DETR [13]	ResNet-50	ECCV20	65.4	70.4	79.5	70.5	40.2	75.7	58.4	68.9	58.5	67.4	70.8	58.4	49.7	74.7	68.4	70.9	76.3	56.9	65.3	62.7	65.5
‡ DAB-DETR [38]	ResNet-50	ICLR22	38.0	57.7	62.5	72.0	22.1	72.2	43.1	53.5	44.4	62.3	48.9	42.3	43.9	12.0	44.0	17.0	58.9	47.0	13.4	45.1	45.0
‡ Deformable-DETR [16]	ResNet-50	ICLR21	70.9	55.8	65.0	82.7	23.4	75.9	37.1	37.8	45.4	55.1	56.2	09.7	44.9	55.8	38.0	44.4	79.7	36.6	36.1	69.5	51.0
‡ DINO [36]	ResNet-50	ICLR23	47.8	71.5	62.9	78.7	28.7	72.7	53.3	55.9	45.3	67.3	55.9	53.9	49.3	16.9	48.6	32.0	71.0	52.4	17.1	52.8	51.7
Large-scale object detector																							
ASSD [71]	VGG-16	TGRS22	85.6	82.4	75.8	89.5	40.7	77.6	64.7	67.1	61.7	80.8	78.6	62.0	58.0	84.9	76.7	65.3	87.9	62.4	44.5	76.3	71.1
‡ TTRD-DA [72]	ResNet-50	REMS22	77.9	80.5	70.1	86.3	39.7	77.9	59.3	59.0	54.4	74.6	73.9	49.2	57.8	74.2	61.1	69.8	84.0	58.8	50.5	77.2	66.8
‡ DIAGTR-Detor [73]	DIAG-TR	JSTARS22	68.4	72.2	71.9	85.1	29.8	76.8	60.2	70.3	**73.2**	75.1	35.5	35.5	50.8	41.6	71.8	32.9	81.0	53.1	33.8	78.6	72.0
SDPNet [74]	CSPDarkNet-53	TGRS23	86.8	81.4	83.4	**90.5**	**52.7**	78.1	63.2	**87.8**	**73.6**	**81.9**	**85.6**	65.2	**64.7**	80.4	**82.6**	77.8	87.5	56.1	59.9	84.7	**76.2**
‡ CI_DETR [40]	ResNet-50	JSTARS23	80.3	74.3	71.7	89.1	43.9	78.6	60.2	60.3	62.6	75.4	74.6	61.8	57.3	89.5	54.2	74.5	85.4	57.7	56.1	81.6	69.5
AGMF-Net [75]	GIEM	TGRS24	**90.9**	72.8	79.3	**89.7**	44.8	81.4	59.3	66.0	62.9	73.9	79.3	65.0	61.7	**91.7**	78.6	75.8	**90.7**	60.0	58.1	83.1	73.2
‡ GOCPK [42]	ResNet-50	JSTARS25	75.6	81.1	71.9	86.8	43.8	79.4	**74.0**	59.6	58.4	80.0	76.2	60.2	58.0	83.2	60.0	70.5	87.1	63.1	54.7	80.1	70.2
Tiny object detector																							
ORFENet [24]	ResNet-50	TGRS24	76.8	77.6	85.7	74.3	44.6	83.7	62.2	73.9	64.5	70.2	73.2	66.5	56.7	83.2	78.7	77.5	83.8	61.4	70.4	71.9	71.8
KLDet [11]	ResNet-50	TGRS24	67.5	81.1	75.5	88.2	47.4	78.6	65.0	82.9	72.7	79.9	80.8	57.1	61.2	**91.2**	74.6	75.2	87.4	56.9	56.0	**90.2**	73.5
Libra-SOD [12]	ResNet-50	KBS24	70.3	**84.2**	76.0	88.6	46.9	79.2	66.5	81.6	69.7	**82.4**	**82.6**	62.3	60.8	89.8	75.4	69.5	88.6	61.8	50.3	**88.4**	73.7
‡ DQ-DETR [18]	ResNet-50	ECCV25	82.4	81.3	**90.4**	79.7	49.3	**87.2**	66.9	78.3	69.8	73.4	76.7	**67.9**	58.5	87.4	80.8	**80.7**	86.4	**63.8**	**72.4**	75.3	75.4
DART	DAT		**93.3**	**86.4**	**93.4**	85.2	**51.2**	**91.1**	**70.5**	**83.1**	72.3	78.8	**82.6**	**69.6**	**64.0**	85.9	**84.3**	**86.4**	**91.6**	**64.2**	**75.1**	80.6	**79.5**

TABLE XIII

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR TEST SET. RED DENOTES THE BEST RESULTS, AND BLUE SECOND-BEST. † INDICATES METHODS EMPLOYING SELF-ATTENTION MECHANISM WITHIN FPNS, AND ‡ INDICATES DETR-LIKE METHODS

Method	AP	AP_{50}	AP_{vt}	AP_t
Generic object detector				
YOLOv3 [64]	40.9	64.3	00.1	03.3
Auto Assign [76]	37.8	63.3	00.2	03.0
YOLOX [77]	48.6	72.5	01.1	08.5
‡DETR [13]	40.6	61.5	00.3	02.2
Tiny object detector				
ORFENet [24]	47.4	70.7	00.9	06.5
RFLA [50]	46.5	68.1	00.3	05.1
†CAF^{2}ENet-M [61]	53.4	**75.4**	**01.4**	**08.6**
‡DQ-DETR [18]	**53.7**	75.3	01.3	08.2
DART	**56.2**	**79.5**	**01.8**	**09.5**

both effectiveness and strong generalization capability. Specifically, in terms of class-wise performance, it demonstrates a strong ability to handle various object sizes, achieving the highest AP for larger objects ranging (520 pixels→800 pixels), such as HA, TS, stadium (STA), and APO, medium-sized objects between (160 pixels→400 pixels), including STO, APL, TC, BF, and CH, and small objects like vehicles (VE) at 40 pixels. Furthermore, it effectively tackles interclass similarity and intraclass diversity, accurately differentiating diverse CH instances while recognizing the similarity between DAM and BR objects, achieving the second-best AP for both DAM and BR, respectively. Fig. 12 illustrates several detection results from the test set, highlighting the model's effectiveness in identifying various objects. Table XIII further compares the performance of various tiny-object detection methods on DIOR, illustrating YOLOX's [77] limitations (AP_{vt} 01.1%, AP_t 8.5%) and minor improvements from CAF^{2}ENet (AP_{vt} 01.1–01.4). DQ-DETR [18] shows better results than ORFENet [24] but remains behind the second-best method. In contrast, our DART model excels with an AP of 56.2%, AP_{vt} of 01.8%, and a 0.9% improvement in AP_t compared to CAF^{2}ENet.

V. CONCLUSION

In this article, we analyze that processing entire flattened feature maps in DETR-like methods forces queries to focus on irrelevant regions, creating redundant attention patterns that impair the model's ability to differentiate densely packed or visually similar tiny objects at varying densities. To overcome these challenges, we propose DART, a lightweight Transformer-based-decoder framework for TOD. Specifically, DAT uses ARA to generate dynamic spatial regions that prioritize computational resources on density-aware and semantically rich areas while minimizing focus on irrelevant background regions. Unlike prior DETR-like methods, RAD adopts masked regions from the DAT as input and computes MRA for each query, quantifying the attention a query pays to relevant region features while reducing focus on overlapping or irrelevant areas. Meanwhile, query diversity loss encourages distinct attention patterns by penalizing overlap among regions, enabling queries to focus more effectively on complementary regions within the image. By adapting to data density and directing queries to essential areas within the image, DART significantly reduces the computational burden. Operating at 45 FPS, DART demonstrates state-of-the-art capabilities on the AI-TOD, DOTA-v2.0, and LEVIR-Ship benchmarks, requiring only 13 million parameters and 68 GFLOPs. Furthermore, DART demonstrates strong performance in challenging real-world scenarios, effectively detecting dense clusters of tiny vehicles in parking lots and precisely localizing extremely small ships navigating marine environments under adverse weather conditions and strong sea waves. In addition, its robust generalization capabilities are exhibited on larger objects within the DIOR dataset through the model's adept handling of both interclass similarity and intraclass diversity.

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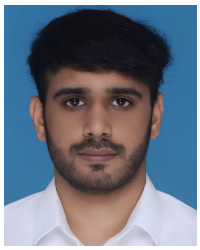
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